

Solange Daniela Correa Holguin

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SOFTWARE & CLOUD ENGINEERING

SOFTWARE ENGINEERING | CLOUD COMPUTING (AWS & AZURE) | FULL-STACK DEVELOPMENT | DATABASE SYSTEMS | AGILE & CI/CD | NETWORKING FUNDAMENTALS

TECHNICAL SKILLS

- **Programming:** Python | C++ | C# | JavaScript | SQL | Java | Bash | PowerShell | HTML | CSS | Linux
- **Frameworks & Tools:** React | Node.js | .NET | WPF | Tkinter | Pandas | XAML | Excel Interop
- **Databases:** MySQL | MongoDB | DynamoDB | SQL Server | ER diagrams | CRUD | Triggers, Normalization
- **Cloud:** AWS (EC2, Lambda, API Gateway, DynamoDB, S3, IAM, Load Balancer) | Azure (VMs, SQL, VPN, SSH tunneling, scaling, performance, monitoring)
- **Networking & Systems:** TCP/IP | UNIX/Linux | Routing & Switching (BGP, OSPF) | Network Troubleshooting | Packet Analysis (Wireshark)
- **Practices:** Agile/Scrum | CI/CD pipelines | SDLC | OOP | MVC | Infrastructure as Code (IaC) | Debugging | Code Reviews | System Monitoring & Troubleshooting
- **Other Tools:** Git | GitHub | MySQL Workbench | Visual Studio | VS Code | PowerShell
- **Soft Skills:** Strong Communication | Team Collaboration | Problem Solving | Time Management | Adaptability | Bilingual (English, Spanish)

WORK EXPERIENCE

Winthrop University - Rock Hill, SC

Aug 2024 – May 2025

Software Engineer | Dowdy Excel Project (Capstone Project - Apprenticeship)

Designed and delivered a **reporting and export system** for a nonprofit client, hosted on **Microsoft Azure**. Built a **Python/Tkinter UI** with a **MySQL backend** to automate data reporting and enable exports into **Excel, Word, and PDF**.

- Built **dynamic filters** (Program Title, Main Category, Subcategory, Date Range) enabling stakeholders to isolate and analyze specific data segments.
- Applied **SDLC principles** (requirements gathering, design, implementation, testing, and deployment) to deliver a secure and maintainable solution.
- Implemented **data validation, SQL injection protection, and error handling** to enhance reporting **security and compliance reliability**.
- Developed export functionality using **Pandas** and **fpdf**, ensuring accurate data transfer into Excel, Word, and PDF reports.
- Configured **Azure-hosted MySQL server** with **SSH tunneling** and migrated production data with schema refactoring.
- Conducted **unit/integration testing, debugging, code reviews**, ensuring functionality and maintainability across the system lifecycle.
- Delivered client demos, Agile sprint updates, and comprehensive documentation to support long-term adoption and maintenance.

Winthrop University - Rock Hill, SC

Jan 2025 – May 2025

Cloud Developer | (AWS Cloud Projects - Apprenticeship)

Designed and deployed cloud-based applications on **Amazon Web Services (AWS)**, focusing on scalability, automation, and secure infrastructure management.

- Deployed a **scalable, load-balanced web application** using **Amazon EC2, Elastic Load Balancer (ALB)**, and **custom security groups** to simulate high availability in production.
- Built a **serverless backend** with **AWS Lambda, API Gateway, and DynamoDB**, supporting **RESTful API** operations without server management.
- Configured and mounted **Amazon EFS** across **EC2 instances**, integrating with **Lambda functions** for shared storage and automated data processing.
- Implemented **least-privilege IAM roles** with custom **JSON policies**; **tested and debugged access controls** across multiple environments (FullAccess, ReadOnlyAccess, Admin).
- Applied **CI/CD principles** with **automated testing and monitoring** using **GitHub workflows** to streamline deployments and ensure reliability.

Carolina IT Solutions, Charlotte, NC

Sept 2024 - Sept 2025

IT Support Technician

Delivered Tier 1 technical support, improving issue resolution time by 25% and increasing user satisfaction.

- Configured **Azure VPN gateways** and deployed **Office 365 accounts**, strengthening secure remote access and productivity.
- Monitored and maintained **Hyper-V VMs** supporting business apps, reducing downtime by 30%.
- Automated repetitive tasks with **PowerShell scripts**, documented issues in a ticketing system, and enhanced reporting accuracy

Winthrop University - Rock Hill, SC

Aug 2024 – Dec 2024

Network Programmer | Reliable Data Transfer (RDT) Protocol Simulator Project

Developed a Python implementation of **rdt3.0 (Alternating Bit Protocol)** to ensure in order, error-free delivery over an unreliable channel.

- Designed **sender (EntityA)** and **receiver (EntityB)** with **checksums, ACK/NACK validation, retransmission timers, and sequence control**.
- Simulated **packet loss, corruption, and network delay**, testing and debugging reliability through automated experiments and performance summaries.
- Applied **TCP/IP reliability principles (flow control, retransmission logic)** within a **UNIX-like simulation environment**, strengthening **transport layer fundamentals**.

EDUCATION

Winthrop University - Rock Hill, SC	Jan 2024 - May 2025
Bachelor of Science - Computer Science, Minor in Mathematics	GPA: 3.67
Inducted Honors: Upsilon Pi Epsilon (UPE) International Honor Society in Computing	
Relevant Coursework: Software Engineering, Networking, Operating Systems, Cloud Computing, C#, C++, Python, Java, JavaScript	
UNIVERSITY OF NORTH CAROLINA - Charlotte, NC	May 2024 - Aug 2024
Summer 2024 Coursework in Computer Science	GPA: 4.00
Relevant Coursework: Introduction to Computer Science II, Intro to Computer Systems, Data Structures & Algorithms	
ESPOL (Higher Polytechnic School of the Litoral) – Guayaquil, Ecuador	Jan 2015 – Jan 2018
Bachelor of Telecommunication and Electronic Engineering (Incomplete – 3 years equivalent per WES)	GPA: 3.70
WES evaluation: Equivalent to three years of U.S. regionally accredited undergraduate study.	
Relevant Coursework: Electrical Network Analysis I & II, Telecommunications Fundamentals, Programming Fundamentals, Programming for Telecommunications, Digital Collaboration Tools	

CERTIFICATIONS

- AWS Certified Cloud Practitioner – In Progress(Expected Dec 2025)